

Multiple Design-and-Simulation Tools in One: **Fedora Electronic Lab**

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Are you working on a new electronics project from scratch? Does it include a designing-and-automating process? Do you have all the required electronics design and automation (EDA) tools for the same? I understand that such projects need a lot of designing tools, and it is tough to find all EDA tools separately and then install these one by one on Windows. So, if you are well-versed with Fedora operating system (OS) and are using one, the solution to all your problems would be Fedora Electronic Laboratory (FEL).

FEL is a spin on Fedora that contains all software one would require for EDA. It provides all the different open source hardware design flows depending on the semiconductor industry's current trends.

Reason for existence

The usual problem that designers have at hand is that they do not find any open source EDA solution for a real-life problem. Now, while engineers use EDA software to design chips or circuit boards, it requires a set of hardware design tools to design chips. However, the same set of hardware design tools need not work with every hardware design project. Although FEL is limited to Fedora OS, it could make its existence possible only because it had the solution to all problems discussed above.

The open source EDA software included in FEL meets the demands of field-proven problems. This open source platform helps you keep in



Fig. 1: Fedora Electronic Laboratory

About Fedora

You might be wondering what Fedora is—whether it is an OS or software package. Let me help clear the confusion. Fedora is a Linux OS, developed by an open source community and owned by Red Hat. It includes a complete set of free and open source tools to support a wide range of operators from students to professionals, and from hobbyists to developers. Fedora supports two primary architectures—Intel x86 and ARM. Although commercial versions of Linux are based on Fedora, Fedora Linux is freely available for use and customisation.

What is a live image. A live image is an ISO file that is used to create live media (CD/DVD/USB) that can be executed without being installed on a hard drive. This means that it can run without having to install additional OS on your computer.

It does not take much time to install and can be configured with an uninterrupted storage. Apart from this, it has other advantages too. It can be used as a back-up to the one installed on your hardware.

Note. The document on how to install FEL has been included in this month's DVD for EFY Plus.

pace with the current technological race with open source community support. This community helps users with installation and use of any EDA tool in the spin.

FEL as a platform has the ability to map the three methodologies—design, simulation and verification. These methodologies enable one to have a better hardware design flow

and experience. These also help Fedora achieve its aim to support innovation and development brought by the open source community.

Collaborative hardware review

The digital hardware design has many requirements, one of which is to track files and feed these into multiple EDA tools. Then, reports and